



15.0.1 UGC NET CSE | December 2007 | Part 2 | Question: 1



A box contains six red balls and four green balls. Four balls are selected at random from the box. What is the probability that two of the selected balls are red and two are green ?

- A. $\frac{3}{7}$ B. $\frac{4}{7}$ C. $\frac{5}{7}$ D. $\frac{6}{7}$

ugcnetcse-dec2007-paper2

Answer key

15.1

Binary Tree (1)

15.1.1 Binary Tree: UGC NET CSE | July 2018 | Part 2 | Question: 25



A text is made up of the characters A, B, C, D, E each occurring with the probability 0.08, 0.40, 0.25, 0.15 and 0.12 respectively. The optimal coding will have the average length of

- A. 2.4 B. 1.87 C. 3.0 D. 2.15

ugcnetcse-july2018-paper2 probability binary-tree data-structures

Answer key

15.2

Joint Distribution (1)

15.2.1 Joint Distribution: UGC NET CSE | December 2018 | Part 2 | Question: 100



A full joint distribution for the Toothache, Cavity and Catch is given in the table below.

	Toothache		$\neg$ Toothache	
	Catch	$\neg$ Catch	Catch	$\neg$ Catch
Cavity	0.108	0.012	0.072	0.008
$\neg$ Cavity	0.016	0.064	0.144	0.576

What is the probability of Cavity, given evidence of Toothache?

- A. $\langle 0.2, 0.8 \rangle$ B. $\langle 0.4, 0.8 \rangle$
 C. $\langle 0.6, 0.8 \rangle$ D. $\langle 0.6, 0.4 \rangle$

ugcnetcse-dec2018-paper2 joint-distribution probability non-gatecse

Answer key

15.3

Poisson Distribution (1)

15.3.1 Poisson Distribution: UGC NET CSE | December 2011 | Part 2 | Question: 27



The multiuser operating system, 20 requests are made to use a particular resource per hour, on an average the probability that no request are made in 45 minutes is

- A. e^{-15} B. e^{-5} C. $1-e^{-5}$ D. $1-e^{-10}$

ugcnetcse-dec2011-paper2 probability poisson-distribution

Answer key

15.4

Probability (9)

15.4.1 Probability: UGC NET CSE | August 2016 | Part 2 | Question: 4



What is the probability that a randomly selected bit string of length 10 is a palindrome ?

A. $\frac{1}{64}$

B. $\frac{1}{32}$

C. $\frac{1}{8}$

D. $\frac{1}{4}$

ugcnetcse-aug2016-paper2 probability

Answer key **15.4.2 Probability: UGC NET CSE | August 2016 | Part 3 | Question: 54**

An experimental file server is up 75% of the time and down for 25% of the time due to bugs. How many times does this file server have to be replicated to give an availability of at least 99% ?

A. 2

B. 4

C. 8

D. 16

ugcnetcse-aug2016-paper3 probability

Answer key **15.4.3 Probability: UGC NET CSE | December 2011 | Part 2 | Question: 49**

What is the probability of choosing correctly an unknown integer between 0 and 9 with 3 chances ?

A. $\frac{963}{1000}$

B. $\frac{973}{1000}$

C. $\frac{983}{1000}$

D. $\frac{953}{1000}$

ugcnetcse-dec2011-paper2 probability

Answer key **15.4.4 Probability: UGC NET CSE | December 2018 | Part 2 | Question: 3**

A box contains six red balls and four green balls. Four balls are selected at random from the box. What is the probability that two of the selected balls will be red and two will be in green?

A. $\frac{1}{14}$

B. $\frac{3}{7}$

C. $\frac{1}{35}$

D. $\frac{1}{9}$

ugcnetcse-dec2018-paper2 probability

Answer key **15.4.5 Probability: UGC NET CSE | July 2016 | Part 2 | Question: 4**

There are three cards in a box. Both sides of one card are black, both sides of one card are red and the third card has one black side and one red side. We pick a card at random and observe only one side. What is the probability that the opposite side is the same colour as the one side we observed?

A. $\frac{3}{4}$

B. $\frac{2}{3}$

C. $\frac{1}{2}$

D. $\frac{1}{3}$

ugcnetcse-july2016-paper2 engineering-mathematics probability

Answer key **15.4.6 Probability: UGC NET CSE | July 2018 | Part 2 | Question: 80**

Consider the set of all possible five-card poker hands dealt fairly from a standard deck of fifty-two cards. How many atomic events are there in the joint probability distribution?

A. 2, 598, 960

B. 3, 468, 960

C. 3, 958, 590

D. 2, 645, 590

ugcnetcse-july2018-paper2 probability

Answer key **15.4.7 Probability: UGC NET CSE | July 2018 | Part 2 | Question: 84**

Digital data received from a sensor can fill up 0 to 32 buffers. Let the sample space be $S = \{0, 1, 2, \dots, 32\}$ where the sample j denote that j of the buffers are full and $p(i) = \frac{1}{562}(33 - i)$. Let A denote the event that the even number of buffers are full. Then $P(A)$ is

A. 0.515

B. 0.785

C. 0.758

D. 0.485

Answer key 

15.4.8 Probability: UGC NET CSE | Junet 2015 | Part 2 | Question: 2



Considering an experiment of tossing two fair dice, one black and one red. What is the probability that the number on the black die divides the number on red die?

- A. $\frac{22}{36}$ B. $\frac{12}{36}$ C. $\frac{14}{36}$ D. $\frac{6}{36}$

Answer key 

15.4.9 Probability: UGC NET CSE | September 2013 | Part 3 | Question: 38



How many people must there be in a room before there is a 50% chance that two of them were born on the same day of the year?

- A. At least 23 B. At least 183 C. At least 366 D. At least 730

Answer key 

Answer Keys

15.0.1	Q-Q	15.1.1	Q-Q	15.2.1	Q-Q	15.3.1	A	15.4.1	Q-Q
15.4.2	Q-Q	15.4.3	Q-Q	15.4.4	Q-Q	15.4.5	B	15.4.6	Q-Q
15.4.7	Q-Q	15.4.8	C	15.4.9	A				